

OSLAB PRACTICAL-1 PROGRAMS & OUTPUT

Date: 05-08-26

Program-1 & OUTPUT:

```
sainath@Sainath:~/practical_1$ cat program1.c
#include <stdio.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    printf("You entered: %s", command);

    return 0;
}
sainath@Sainath:~/practical_1$ ./program1
Enter a Linux command: ls
You entered: ls
```

Program-2 & OUTPUT:

```

sainath@Sainath:~/practical_1$ cat program2.c
#include <stdio.h>
#include <unistd.h>

int main() {

    pid_t pid = fork();

    if(pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
    }
    else if(pid > 0) {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
    }
    else {
        printf("Fork Failed!\n");
    }

    return 0;
}

sainath@Sainath:~/practical_1$ ./program2
Parent Process
Parent PID : 750
Child Process
Child PID : 751

```

Program-3 & OUTPUT:

```
sainath@Sainath:~/practical_1$ cat program3.c
#include <stdio.h>
#include <unistd.h>

int main() {

    printf("Executing 'ls -l' command...\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("This line will execute only if exec fails.\n");

    return 0;
}
sainath@Sainath:~/practical_1$ ./program3
Executing 'ls -l' command...
total 100
-rwxr-xr-x 1 sainath sainath 16104 Aug  5 03:25 program1
-rw-r--r-- 1 sainath sainath   198 Aug  5 03:24 program1.c
-rwxr-xr-x 1 sainath sainath 16088 Aug  5 03:28 program2
-rw-r--r-- 1 sainath sainath   365 Aug  5 03:28 program2.c
-rwxr-xr-x 1 sainath sainath 16008 Aug  5 03:29 program3
-rw-r--r-- 1 sainath sainath   219 Aug  5 03:29 program3.c
-rwxr-xr-x 1 sainath sainath 16088 Aug  5 03:30 program4
-rw-r--r-- 1 sainath sainath   388 Aug  5 03:30 program4.c
-rwxr-xr-x 1 sainath sainath 16136 Aug  5 03:31 program5
-rw-r--r-- 1 sainath sainath   405 Aug  5 03:31 program5.c
```

Program-4 & OUTPUT:

```
sainath@Sainath:~/practical_1$ cat program4.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid = fork();

    if(pid == 0) {
        printf("Child Process Running...\n");
        sleep(3);
        printf("Child Process Finished.\n");
    }
    else {
        printf("Parent Waiting...\n");
        wait(NULL);
        printf("Parent Resumed After Child Completion.\n");
    }

    return 0;
}
sainath@Sainath:~/practical_1$ ./program4
Parent Waiting...
Child Process Running...
Child Process Finished.
Parent Resumed After Child Completion.
```

Program-5 & OUTPUT:

```
sainath@Sainath:~/practical_1$ cat program5.c
#include <stdio.h>
#include <unistd.h>

int main() {

    pid_t pid = fork();

    if(pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
    }
    else if(pid > 0) {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }

    return 0;
}

sainath@Sainath:~/practical_1$ ./program5
Parent Process
Parent PID : 758
Child PID : 759
Child Process
Child PID : 759
Parent PID : 380
```

COMMANDS:

uname -a:

```
sainath@Sainath:~/practical_1$ uname -a
Linux Sainath 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

lscpu:

```
sainath@sainath:~/practical_1$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 12
On-line CPU(s) list: 0-11
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-13420H
CPU family: 6
Model: 186
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 2
BogoMIPS: 5222.40
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 288 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-11
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
```

lsblk:

```
sainath@sainath:~/practical_1$ lsblk
NAME                                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                                8:0    0 356.9M  1 disk
sdb                                8:16    0 159.4M  1 disk
sdc                                8:32    0      2G   0 disk [SWAP]
sdd                                8:48    0      1T   0 disk /snap
                                   /mnt/wslg/distro
                                   /
```

ps:

```
sainath@Sainath:~/practical_1$ ps
  PID TTY          TIME CMD
  385 pts/0        00:00:00 bash
  764 pts/0        00:00:00 ps
```

top:

```
sainath@Sainath:~/practical_1$ top
top - 03:40:07 up 43 min,  1 user,  load average: 0.02, 0.03, 0.00
Tasks:  79 total,   1 running,  78 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.7 us,  0.0 sy,  0.0 ni, 98.5 id,  0.7 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7762.6 total,  6997.0 free,   567.7 used,   350.9 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  7195.0 avail Mem

   PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
    765 sainath   20   0    9300    5432    3256  R   10.0   0.1   0:00.01 top
      1 root        20   0   21732   13112   9732  S    0.0   0.2   0:01.63 systemd
      2 root        20   0    3180    2200    2072  S    0.0   0.0   0:00.01 init-systemd(Ub
      7 root        20   0    3196    2128    2008  S    0.0   0.0   0:00.00 init
     59 root       19  -1   50356   16412   15280  S    0.0   0.2   0:00.36 systemd-journal
    106 root        20   0   25536    6812    5080  S    0.0   0.1   0:02.43 systemd-udevd
    115 root        20   0   152944   1776    1520  S    0.0   0.0   0:00.00 snapfuse
    120 root        20   0   153076   1836    1512  S    0.0   0.0   0:00.00 snapfuse
    126 root        20   0   153076   1684    1432  S    0.0   0.0   0:00.00 snapfuse
    131 root        20   0   153076   1684    1424  S    0.0   0.0   0:00.00 snapfuse
    135 root        20   0    25116   4896    3564  S    0.0   0.1   0:00.05 (udev-worker)
    136 root        20   0    25012   4340    3020  S    0.0   0.1   0:00.01 (udev-worker)
    137 root        20   0    25124   5136    3792  S    0.0   0.1   0:00.01 (udev-worker)
    138 root        20   0    25124   3884    2540  S    0.0   0.0   0:00.00 (udev-worker)
    139 root        20   0    25556   6516   5140  S    0.0   0.1   0:00.02 (udev-worker)
    140 root        20   0    25132   5068    3724  S    0.0   0.1   0:00.40 (udev-worker)
    141 root        20   0    25136   4724    3372  S    0.0   0.1   0:00.00 (udev-worker)
    143 root        20   0    25140   3892    2536  S    0.0   0.0   0:00.00 (udev-worker)
    145 root        20   0    25144   4536    3148  S    0.0   0.1   0:00.00 (udev-worker)
    146 root        20   0    25144   4712    3324  S    0.0   0.1   0:00.00 (udev-worker)
    148 root        20   0   153076   1628    1372  S    0.0   0.0   0:00.00 snapfuse
    151 root        20   0    25144   4500    3072  S    0.0   0.1   0:00.00 (udev-worker)
    152 root        20   0    25144   4296    2876  S    0.0   0.1   0:00.00 (udev-worker)
    153 root        20   0    25252   4704    3216  S    0.0   0.1   0:00.01 (udev-worker)
    154 root        20   0    25144   4372    2936  S    0.0   0.1   0:00.00 (udev-worker)
    155 root        20   0    25144   4644    3200  S    0.0   0.1   0:00.00 (udev-worker)
    158 root        20   0    25144   4284    2864  S    0.0   0.1   0:00.01 (udev-worker)
    159 root        20   0    25144   4200    2780  S    0.0   0.1   0:00.02 (udev-worker)
    160 root        20   0    25300   5252    3776  S    0.0   0.1   0:02.18 (udev-worker)
    161 root        20   0    25144   4336    2908  S    0.0   0.1   0:00.00 (udev-worker)
    162 root        20   0    25276   4500    3020  S    0.0   0.1   0:00.00 (udev-worker)
    163 root        20   0    25276   4352    2884  S    0.0   0.1   0:00.44 (udev-worker)
    164 root        20   0    25276   4652    3172  S    0.0   0.1   0:00.00 (udev-worker)
    165 root        20   0    25276   3752    2276  S    0.0   0.0   0:00.00 (udev-worker)
```

history:


```
sainath@Sainath:~/practical_1$ history
 1  nano demo.c
 2  gcc demo.c -o demo
 3  sudo apt update
 4  gcc demo.c -o demo
 5  sudo apt install build-essential
 6  gcc demo.c -o demo
 7  mkdir OS programs
 8  mkdir OSLAB
 9  cd OSLAB
10  mkdir fork
11  cd fork
12  gedit program1.c
13  apt update
14  apt install gedit
15  gedit --version
16  sudo snap install gedit
17  sudo apt install gedit
18  gedit program1.c
19  gcc program1.c -o process
20  ./program1
21  ./process
22  gedit program2.c
23  gcc program2.c -o process
24  ./process
25  gedit program3.c
26  sudo apt install gedit
27  gcc program3.c -o process
28  ./process
29  gedit program4.c
30  gcc program4.c -o process
31  ./process
32  gedit program5.c
33  gcc program5.c -o process
34  ./process
35  gedit program6.c
36  gcc program6.c -o process
37  ./process
38  gedit program7.c
39  gcc program7.c -o process
40  ./process
41  gedit program10.c
42  gcc program10.c -o process
43  ./process
44  gedit program9.c
```

```
45 gcc program9.c -o process
46 ./process
47 pwd
48 ~/ OSLAB
49 cd ~/OSLAB
50 ls
51 cd ~/fork
52 ls
53 cat program1.c
54 pwd
55 cd ~/ OSLAB
56 cd ~/OSLAB
57 LS
58 ls
59 cd ~/program1.c
60 cd ~/OSLAB
61 cd fork
62 ls
63 cd program1.c
64 cat program1.c
65 ./program1
66 ./program1.c
67 ./process
68 cd ~/OSLAB
69 cd fork
70 ls
71 cat program1.c
72 ./program1.c
73 ./program1
74 ./program1.c
75 ls
76 pwd
77 cd OSLAB
78 mkdir exec
79 cd exec
80 geddit program1.c
81 gedit program1.c
82 gcc program1.c
83 gcc program1.c -o program1.c
84 gcc program1.c -o program1
85 ./program1
86 gedit program2.c
87 gcc program2.c -o program2
88 ./program2
89 gedit program3.c
```

```
90 gcc program3.c -o program3
91 ./program3
92 gedit program4.c
93 gcc program4.c -o program4
94 ./program4
95 gedit program5.c
96 gcc program5.c -o program5
97 ./program5
98 cat program1.c
99 ./program1
100 cat program2.c
101 ./program2
102 cat program3.c
103 ./program3
104 cat program4.c
105 ./program4
106 cat program5.c
107 ./program5
108 cd fork
109 cd ~/OSLAB
110 cd fork
111 cat program1.c
112 ./program1
113 ./program1.c
114 cd ~/OSLAB
115 cd fork
116 cat fork
117 cd ~/OSLAB
118 cd fork
119 cat program1.c
120 gcc program1.c -o program1
121 ls -l
122 ./program1
123 cd ~/OSLAB
124 cd fork
125 cat program1.c
126 gcc program1.c -o program1
127 ./program1.c
128 ls -l
129 ./program1.c
130 ./program1
131 cd ~/OSLAB
132 cd fork
133 cat program1.c
134 gcc program1.c -o program1
```

```
135 ./program1
136 cat program2.c
137 ./program2
138 ./program2.c
139 cd ~/OSLAB
140 cd fork
141 cat program1.c
142 gcc program1.c -o program1
143 ./program1
144 cat program2.c
145 gcc program2.c -o program2
146 ./program2
147 cat program3.c
148 gcc program3.c -o program3
149 ./program3
150 cat program4.c
151 gcc program4.c -o program4
152 ./program4
153 cat program5.c
154 gcc program5.c -o program5
155 ./program5
156 cat program6.c
157 gcc program6.c -o program6
158 ./program6
159 cat program7.c
160 gcc program7.c -o program7
161 ./program7
162 cat program8.c
163 cat program9.c
164 gcc program9.c -o program9
165 ./program9
166 cat program10.c
167 gcc program10.c -o program10
168 ./program10
169 mkdir practical_1
170 cd practical_1
171 gedit program1.c
172 gcc promgram1.c -o program1
173 gcc program1.c -o program1
174 gcc program1.c -o program1
175 ./program1
176 gedit program2.c
177 gcc program2.c -o program2
178 ./program2
179 gedit program3.c
```

```
180 gcc program3.c -o program3
181 ./program3
182 gedit program4.c
183 gcc program4.c -o program4
184 ./program4
185 gedit program5.c
186 gcc program5.c -o program5
187 ./program5
188 cat program1.c
189 ./program1.c
190 ./program1
191 cat program1.c
192 ./program1
193 cat program2.c
194 ./program2
195 cat program3.c
196 ./program3
197 cat program4.c
198 ./program4
199 cat program5.c
200 ./program5
201 uname -a
202 lscpu
203 lsblk
204 ps
205 top
206 history
```